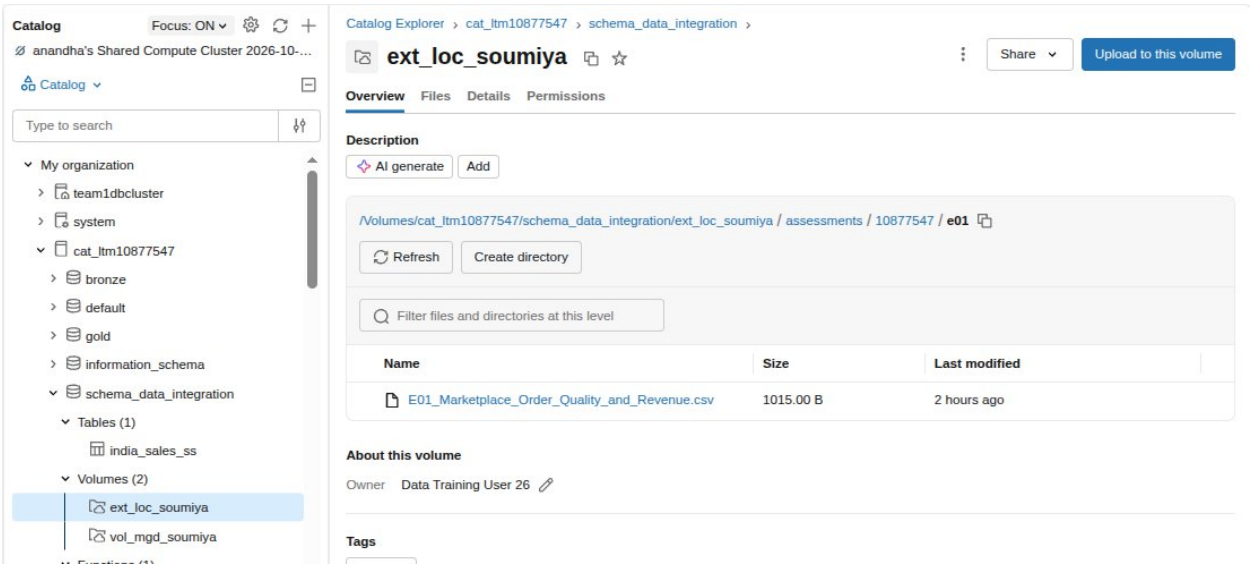
Soumiya_Ayyappan_w9_D5

Easy qn

Step 1:



✓ 11:29 AM (6s) 1

```
%sql
use catalog cat_ltm10877547;
use schema schema_data_integration;
show schemas in cat_ltm10877547;
```

> See performance (3)

Table +

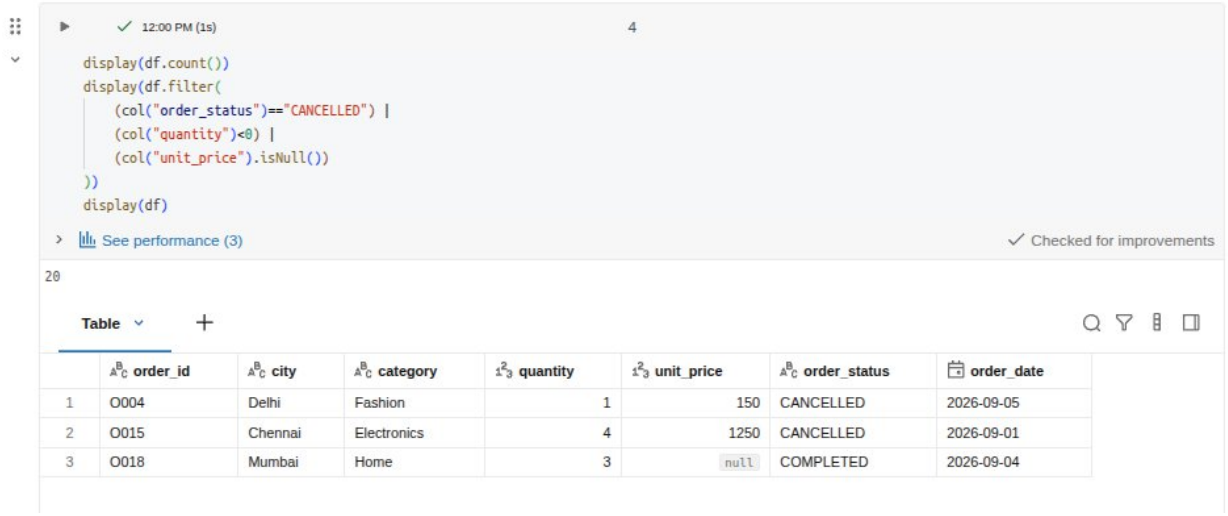
	databaseName
1	bronze
2	default
3	gold
4	information_schema
5	schema_data_integration
6	silver
7	training_assessment_delivery_ops
8	training_assessment_hospitality

8 rows | 5.559s runtime

✓ 12:00 PM (<1s) 3

```
from pyspark.sql.functions import *
from pyspark.sql.types import *
schema=StructType([
    StructField("order_id",StringType(),True),
    StructField("city",StringType(),True),
    StructField("category",StringType(),True),
    StructField("quantity",IntegerType(),True),
    StructField("unit_price",IntegerType(),True),
    StructField("order_status",StringType(),True),
    StructField("order_date",DateType(),True)
])
df=spark.read.csv("/Volumes/cat_ltm10877547/schema_data_integration/ext_loc_soumiya/assessments/10877547/e01/
E01_Marketplace_Order_Quality_and_Revenue.csv",header=True,schema=schema)
```

> df: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, city: string ... 5 more fields]



```
display(df.count())
display(df.filter(
    (col("order_status")=="CANCELLED") |
    (col("quantity")<0) |
    (col("unit_price").isNull())
))
display(df)
```

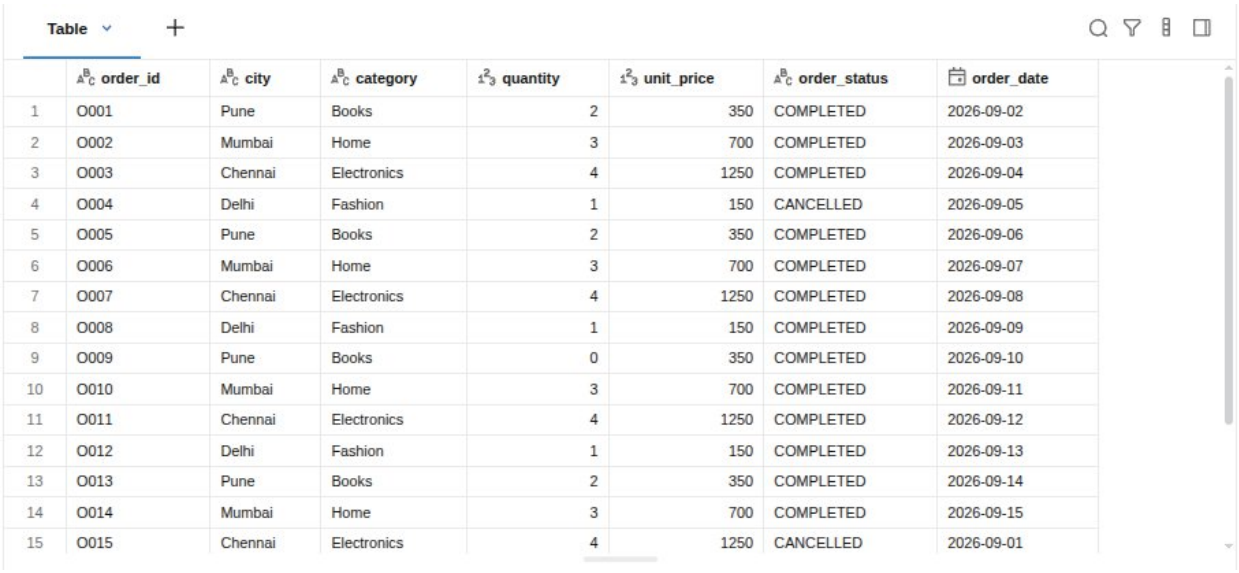
12:00 PM (1s) 4

See performance (3) Checked for improvements

20

Table

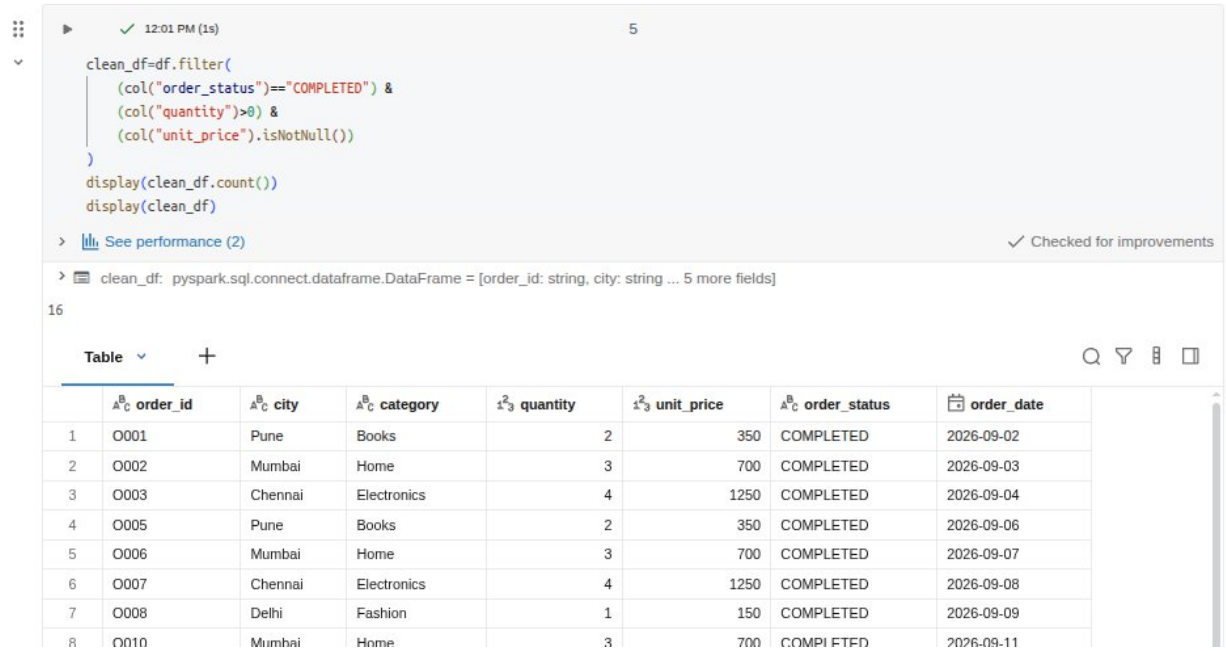
	order_id	city	category	quantity	unit_price	order_status	order_date
1	O004	Delhi	Fashion	1	150	CANCELLED	2026-09-05
2	O015	Chennai	Electronics	4	1250	CANCELLED	2026-09-01
3	O018	Mumbai	Home	3	null	COMPLETED	2026-09-04



Table

	order_id	city	category	quantity	unit_price	order_status	order_date
1	O001	Pune	Books	2	350	COMPLETED	2026-09-02
2	O002	Mumbai	Home	3	700	COMPLETED	2026-09-03
3	O003	Chennai	Electronics	4	1250	COMPLETED	2026-09-04
4	O004	Delhi	Fashion	1	150	CANCELLED	2026-09-05
5	O005	Pune	Books	2	350	COMPLETED	2026-09-06
6	O006	Mumbai	Home	3	700	COMPLETED	2026-09-07
7	O007	Chennai	Electronics	4	1250	COMPLETED	2026-09-08
8	O008	Delhi	Fashion	1	150	COMPLETED	2026-09-09
9	O009	Pune	Books	0	350	COMPLETED	2026-09-10
10	O010	Mumbai	Home	3	700	COMPLETED	2026-09-11
11	O011	Chennai	Electronics	4	1250	COMPLETED	2026-09-12
12	O012	Delhi	Fashion	1	150	COMPLETED	2026-09-13
13	O013	Pune	Books	2	350	COMPLETED	2026-09-14
14	O014	Mumbai	Home	3	700	COMPLETED	2026-09-15
15	O015	Chennai	Electronics	4	1250	CANCELLED	2026-09-01

Step 2:



```
clean_df=df.filter(
    (col("order_status")=="COMPLETED") &
    (col("quantity")>0) &
    (col("unit_price").isNotNull())
)
display(clean_df.count())
display(clean_df)
```

12:01 PM (1s) 5

See performance (2) Checked for improvements

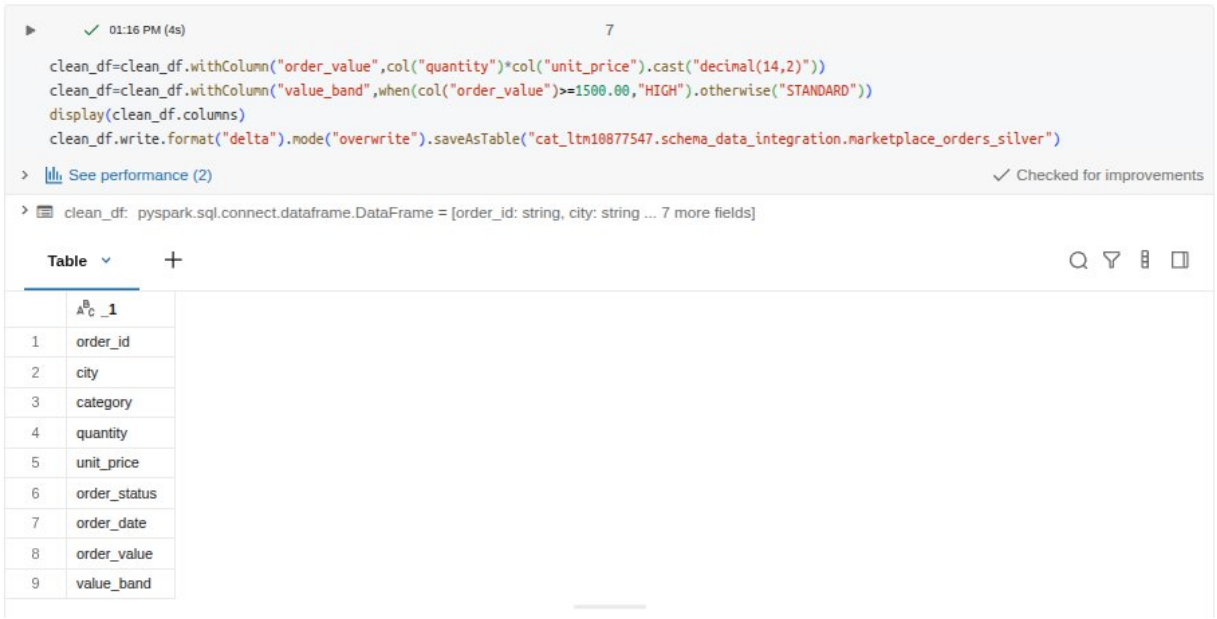
clean_df: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, city: string ... 5 more fields]

16

Table

	order_id	city	category	quantity	unit_price	order_status	order_date
1	O001	Pune	Books	2	350	COMPLETED	2026-09-02
2	O002	Mumbai	Home	3	700	COMPLETED	2026-09-03
3	O003	Chennai	Electronics	4	1250	COMPLETED	2026-09-04
4	O005	Pune	Books	2	350	COMPLETED	2026-09-06
5	O006	Mumbai	Home	3	700	COMPLETED	2026-09-07
6	O007	Chennai	Electronics	4	1250	COMPLETED	2026-09-08
7	O008	Delhi	Fashion	1	150	COMPLETED	2026-09-09
8	O010	Mumbai	Home	3	700	COMPLETED	2026-09-11

Step 3:



```
clean_df=clean_df.withColumn("order_value",col("quantity")*col("unit_price").cast("decimal(14,2)"))
clean_df=clean_df.withColumn("value_band",when(col("order_value")>=1500.00,"HIGH").otherwise("STANDARD"))
display(clean_df.columns)
clean_df.write.format("delta").mode("overwrite").saveAsTable("cat_ltm10877547.schema_data_integration.marketplace_orders_silver")
```

01:16 PM (4s) 7

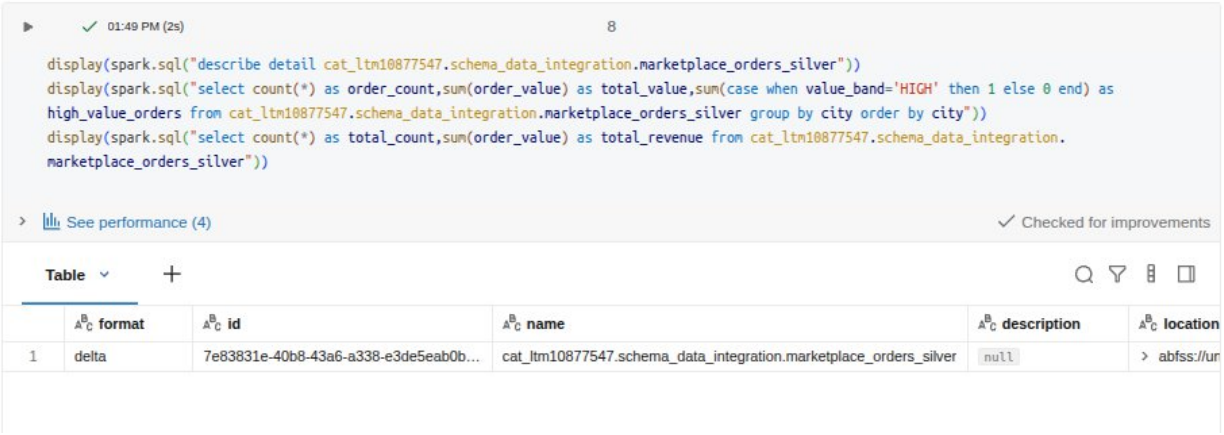
See performance (2) Checked for improvements

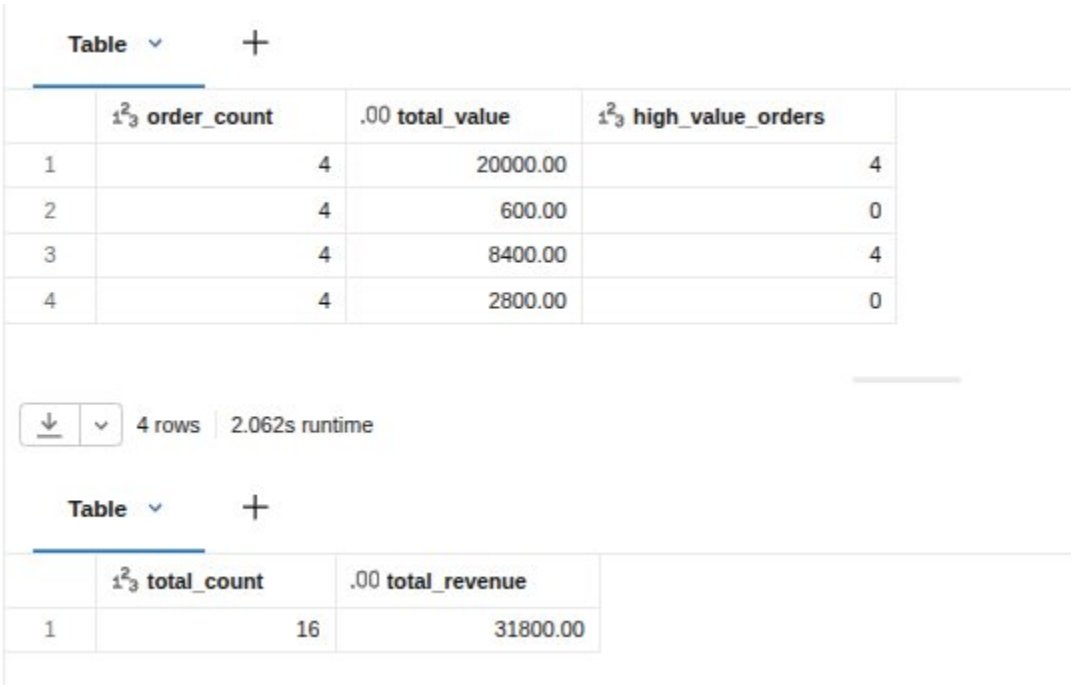
clean_df: pyspark.sql.connect.dataframe.DataFrame = [order_id: string, city: string ... 7 more fields]

Table

	_1
1	order_id
2	city
3	category
4	quantity
5	unit_price
6	order_status
7	order_date
8	order_value
9	value_band

Step 4:





Medium qn

Step 1:

Catalog Explorer › cat_ltm10877547 › schema_data_integration ›

ext_loc_soumiya

Share | Upload to this volume

Overview | Files | Details | Permissions

Description

AI generate | Add

/Volumes/cat_ltm10877547/schema_data_integration/ext_loc_soumiya / assessments / 10877547 / **m01**

Refresh | Create directory

Filter files and directories at this level

Name	Size	Last modified
M01_Industrial_Asset_Delta_Maintenance.csv	492.00 B	21 hours ago

About this volume

Owner Data Training User 26

21 hours ago (10s)

```
%sql
use catalog cat_ltm10877547;
use schema schema_data_integration;
show schemas in cat_ltm10877547;
```

See performance (3)

Table +

	databaseName
1	bronze
2	default
3	gold
4	information_schema
5	schema_data_integration
6	silver
7	training_assessment_delivery_ops
8	training_assessment_hospitality

8 rows | 10.124s runtime

Step 2:

```
11:08 AM (4s)                                                    2

from pyspark.sql.functions import *
from pyspark.sql.types import *
schema=StructType([
 StructField("asset_id",StringType(),True),
 StructField("region",StringType(),True),
 StructField("asset_type",StringType(),True),
 StructField("quantity",IntegerType(),True),
 StructField("unit_cost",DecimalType(12,2),True),
 StructField("total_cost",DecimalType(14,2),True),
 StructField("condition",StringType(),True)
])
df=spark.read.csv("/Volumes/cat_ltm10877547/schema_data_integration/ext_loc_soumiya/assessments/10877547/m01/
M01_Industrial_Asset_Delta_Maintenance.csv",header=True,schema=schema)

df:  pyspark.sql.connect.dataframe.DataFrame = [asset_id: string, region: string ... 5 more fields]
```

```
11:09 AM (3s)                                                    3

df.write.format("delta").mode("overwrite").option("overwriteSchema", "true").saveAsTable("cat_ltm10877547.schema_data_integration.asset_costs")

See performance (1)                                  Checked for improvements
```

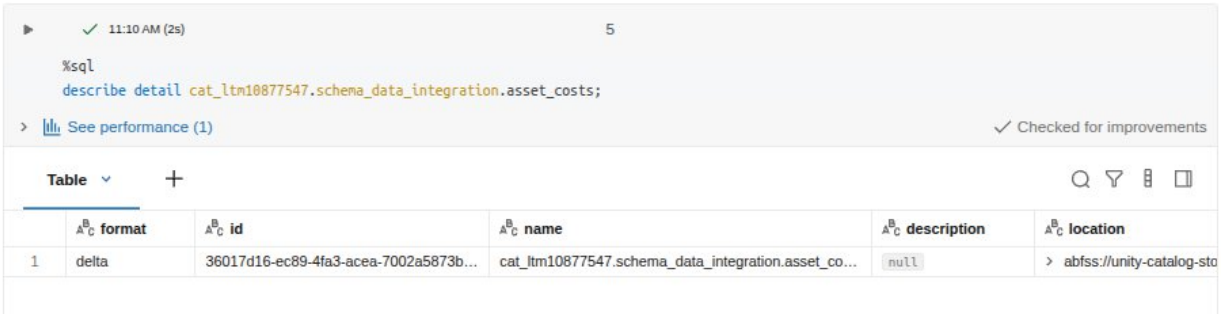
```
11:09 AM (5s)                                                    4

%sql
select count(*),sum(total_cost) from cat_ltm10877547.schema_data_integration.asset_costs ;

See performance (1)
```

Table +

	count(*)	sum(total_cost)
1	12	32400.00



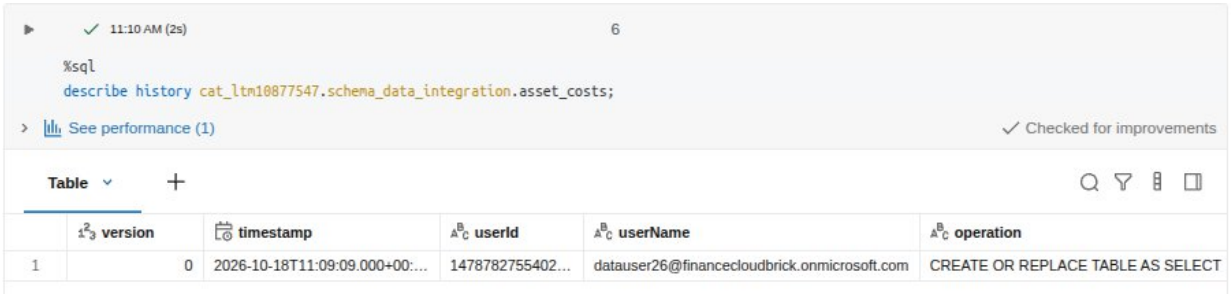
```
%sql
describe detail cat_ltm10877547.schema_data_integration.asset_costs;
```

11:10 AM (2s) 5

See performance (1) Checked for improvements

Table

	format	id	name	description	location
1	delta	36017d16-ec89-4fa3-acea-7002a5873b…	cat_ltm10877547.schema_data_integration.asset_co…	null	abfss://unity-catalog-sto



```
%sql
describe history cat_ltm10877547.schema_data_integration.asset_costs;
```

11:10 AM (2s) 6

See performance (1) Checked for improvements

Table

	version	timestamp	userId	userName	operation
1	0	2026-10-18T11:09:09.000+00:…	1478782755402…	datauser26@financecloudbrick.onmicrosoft.com	CREATE OR REPLACE TABLE AS SELECT

Step 3:



```
#display(spark.sql("select sum(total_cost) from sa_catalog.hr_schema.asset_costs"))
display(spark.sql("update cat_ltm10877547.schema_data_integration.asset_costs set quantity=4 ,unit_cost=650.00,total_cost=2600.00 where
asset_id='A002'"))
display(spark.sql("update cat_ltm10877547.schema_data_integration.asset_costs set quantity=6 ,unit_cost=850.00,total_cost=5100.00 where
asset_id='A007'"))
display(spark.sql("select * from cat_ltm10877547.schema_data_integration.asset_costs"))
display(spark.sql("select sum(total_cost) from cat_ltm10877547.schema_data_integration.asset_costs"))
```

11:11 AM (7s) 8

See performance (6) Optimize

Table

	num_affected_rows
1	1

Table

	num_affected_rows
1	1

1 row | 7.328s runtime

Refreshed at 10/18/26, 11:11 AM (28m ago)

Table

	asset_id	region	asset_type	quantity	unit_cost	total_cost	condition
1	A001	North	Pump	2	400.00	800.00	ACTIVE
2	A003	West	Valve	4	800.00	3200.00	ACTIVE
3	A004	North	Pump	5	1000.00	5000.00	ACTIVE
4	A005	South	Motor	2	400.00	800.00	ACTIVE
5	A006	West	Valve	3	600.00	1800.00	ACTIVE
6	A008	South	Motor	5	1000.00	5000.00	ACTIVE
7	A009	West	Valve	2	400.00	800.00	ACTIVE
8	A010	North	Pump	3	600.00	1800.00	ACTIVE
9	A011	South	Motor	4	800.00	3200.00	ACTIVE
10	A012	West	Valve	5	1000.00	5000.00	ACTIVE
11	A002	South	Motor	4	650.00	2600.00	ACTIVE
12	A007	North	Pump	6	850.00	5100.00	ACTIVE

12 rows | 7.328s runtime

Refreshed at 10/18/26, 11:11 AM (28m ago)

Table

	sum(total_cost)
1	35100.00

Step 4:

✓ 11:13 AM (4s) 10

```
display(spark.sql("delete from cat_ltm10877547.schema_data_integration.asset_costs where asset_id='A006'"))
display(spark.sql("select count(*),sum(total_cost) from cat_ltm10877547.schema_data_integration.asset_costs"))
display(spark.sql("describe history cat_ltm10877547.schema_data_integration.asset_costs"))
```

See performance (4) Optimize

Table

	num_affected_rows
1	1

1 row | 3.528s runtime Refreshed at 10/18/26, 11:13 AM (28m ago)

Table

	count(*)	sum(total_cost)
1	11	33300.00

Table

	version	timestamp	userId	userName	operation
1	4	2026-10-18T11:13:34.000+00:...	1478782755402...	datauser26@financecloudbrick.onmicrosoft.com	DELETE
2	3	2026-10-18T11:11:40.000+00:...	1478782755402...	datauser26@financecloudbrick.onmicrosoft.com	OPTIMIZE
3	2	2026-10-18T11:11:39.000+00:...	1478782755402...	datauser26@financecloudbrick.onmicrosoft.com	UPDATE
4	1	2026-10-18T11:11:36.000+00:...	1478782755402...	datauser26@financecloudbrick.onmicrosoft.com	UPDATE
5	0	2026-10-18T11:09:09.000+00:...	1478782755402...	datauser26@financecloudbrick.onmicrosoft.com	CREATE OR REPLACE TABLE AS SELEC

5 rows | 3.528s runtime Refreshed at 10/18/26, 11:13 AM (29m ago)

✓ 11:14 AM (1s) 11

```
display(spark.sql("select  count(*),sum(total_cost) from cat_ltm10877547.schema_data_integration.asset_costs version as of 0"))
```

See performance (1) ✓ Checked for improvements

Table

	count(*)	sum(total_cost)
1	12	32400.00

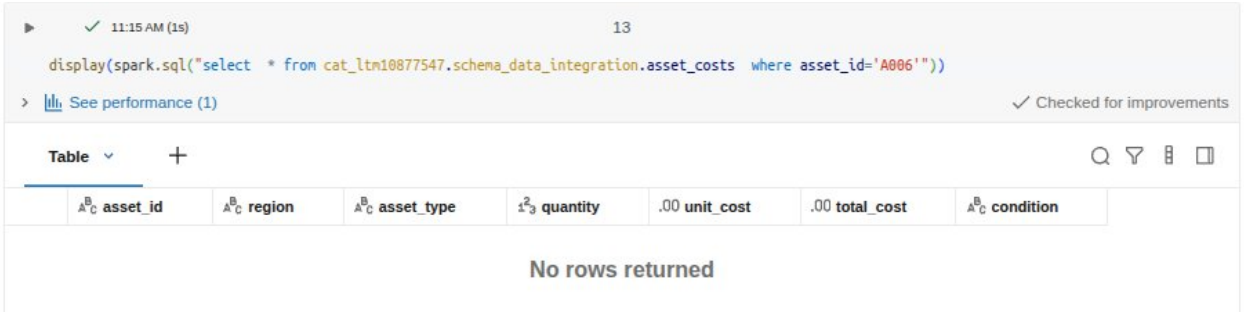
✓ 11:15 AM (1s) 12

```
display(spark.sql("select  * from cat_ltm10877547.schema_data_integration.asset_costs  version as of 0 where asset_id='A006'"))
```

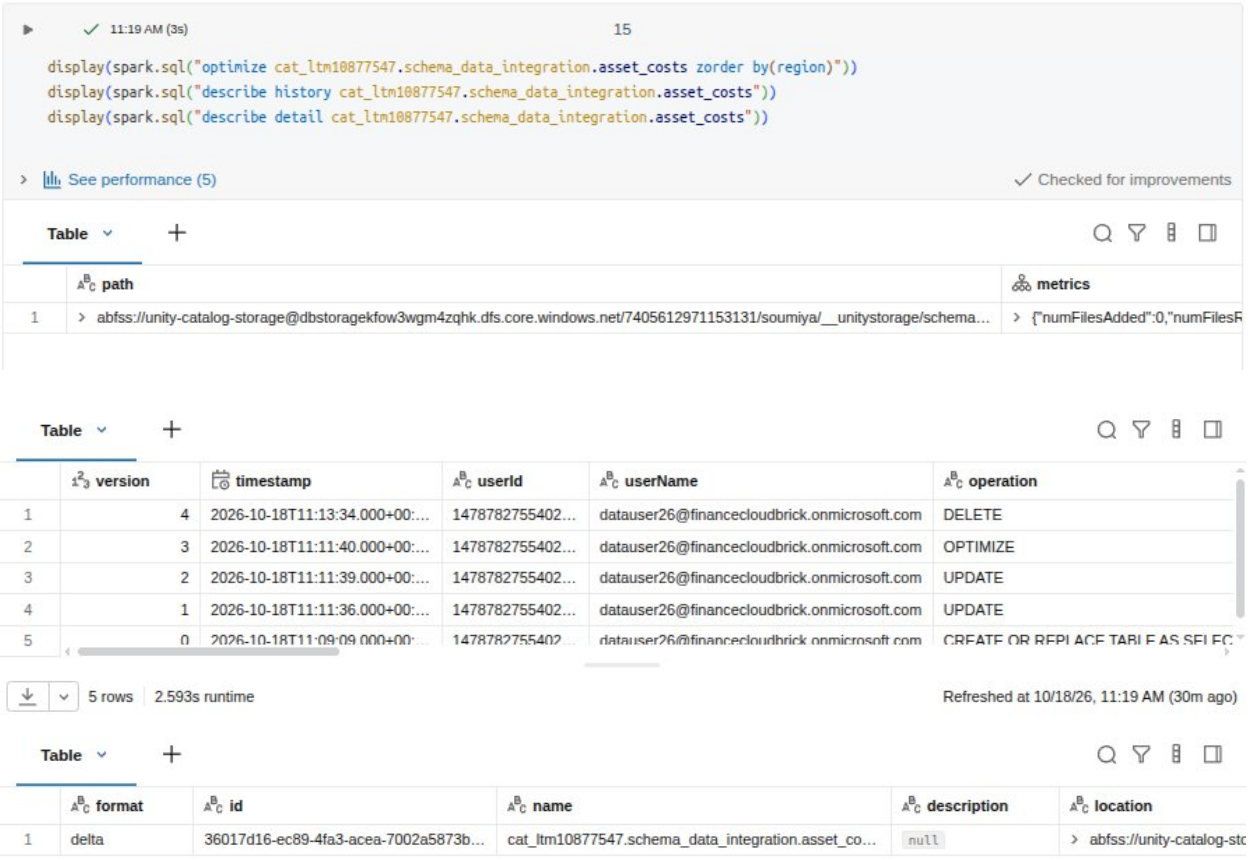
See performance (1) ✓ Checked for improvements

Table

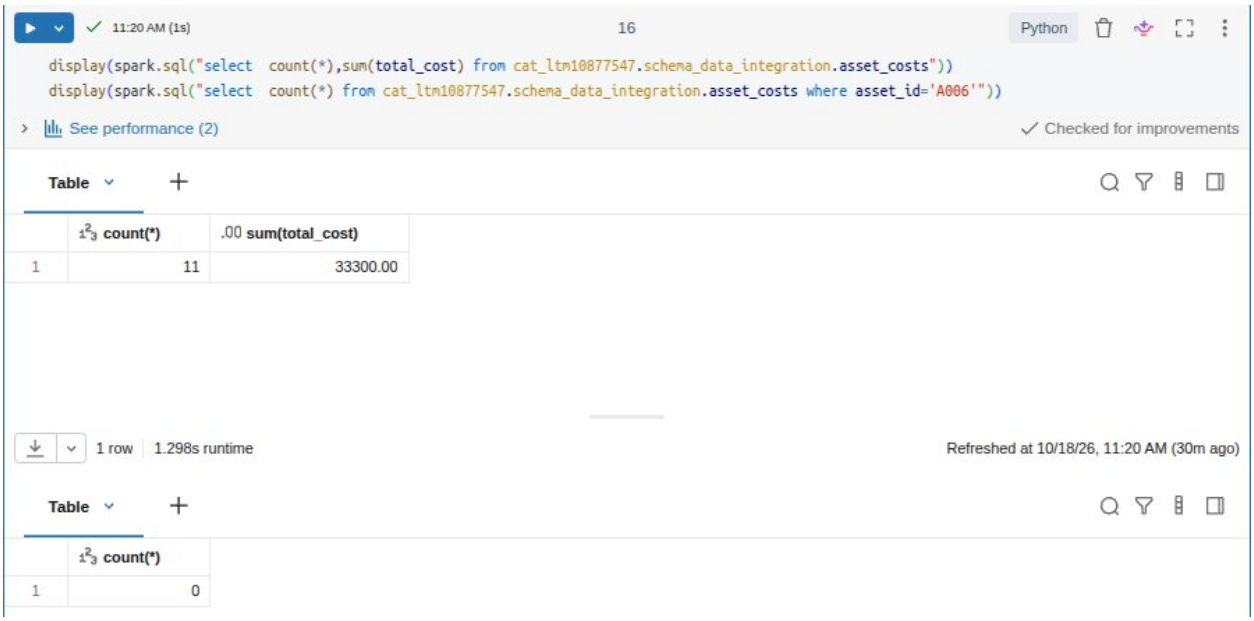
	asset_id	region	asset_type	quantity	unit_cost	total_cost	condition
1	A006	West	Valve	3	600.00	1800.00	ACTIVE


11:15 AM (1s)
13
display(spark.sql("select * from cat_ltm10877547.schema_data_integration.asset_costs where asset_id='A006'"))
See performance (1)
Checked for improvements
Table
asset_id
region
asset_type
quantity
unit_cost
total_cost
condition
No rows returned

Step 5:


11:19 AM (3s)
15
display(spark.sql("optimize cat_ltm10877547.schema_data_integration.asset_costs zorder by(region)"))
display(spark.sql("describe history cat_ltm10877547.schema_data_integration.asset_costs"))
display(spark.sql("describe detail cat_ltm10877547.schema_data_integration.asset_costs"))
See performance (5)
Checked for improvements
Table
path
metrics
1
abfss://unity-catalog-storage@dbstoragekfow3wgm4zqhk.dfs.core.windows.net/7405612971153131/soumiya/__unitystorage/schema...
{"numFilesAdded":0,"numFilesR
Table
version
timestamp
userId
userName
operation
1 4 2026-10-18T11:13:34.000+00:... 1478782755402... datauser26@financecloudbrick.onmicrosoft.com DELETE
2 3 2026-10-18T11:11:40.000+00:... 1478782755402... datauser26@financecloudbrick.onmicrosoft.com OPTIMIZE
3 2 2026-10-18T11:11:39.000+00:... 1478782755402... datauser26@financecloudbrick.onmicrosoft.com UPDATE
4 1 2026-10-18T11:11:36.000+00:... 1478782755402... datauser26@financecloudbrick.onmicrosoft.com UPDATE
5 0 2026-10-18T11:09:09.000+00:... 1478782755402... datauser26@financecloudbrick.onmicrosoft.com CREATE OR REPLACE TABLE AS SELEC
5 rows
2.593s runtime
Refreshed at 10/18/26, 11:19 AM (30m ago)
Table
format
id
name
description
location
1
delta
36017d16-ec89-4fa3-acea-7002a5873b...
cat_ltm10877547.schema_data_integration.asset_co...
null
abfss://unity-catalog-stc

Step 6:


11:20 AM (1s)
16
Python
display(spark.sql("select count(*),sum(total_cost) from cat_ltm10877547.schema_data_integration.asset_costs"))
display(spark.sql("select count(*) from cat_ltm10877547.schema_data_integration.asset_costs where asset_id='A006'"))
See performance (2)
Checked for improvements
Table
count(*)
sum(total_cost)
1
11
33300.00
1 row
1.298s runtime
Refreshed at 10/18/26, 11:20 AM (30m ago)
Table
count(*)
1
0